

Pulse Width Modulation (PWM) on STM32

Course: STM32 Microcontroller

September 19, 2025

Overview

- What is PWM?
- Applications of PWM in embedded systems
- STM32 Timers and PWM generation
- Example: LED brightness, DC motor, servo

What is PWM?

- PWM = Pulse Width Modulation
- A digital signal with fixed frequency but variable duty cycle
- Duty cycle = $\frac{T_{on}}{T_{period}} \times 100\%$
- Controls the average power delivered to a load

PWM Signal

50% duty cycle



75% duty cycle



25% duty cycle



Applications of PWM

- LED dimming
- DC motor speed control
- Servo motor control (1–2 ms pulse)
- Audio signal generation
- Power electronics (inverters, SMPS)

PWM on STM32

- Generated by TIMx (Timer modules)
- Key registers:
 - PSC (Prescaler) – sets clock division
 - ARR (Auto-reload register) – sets period
 - CCRx (Capture/Compare Register) – sets duty cycle
- Output on alternate function pins

Frequency and Duty Cycle

$$f_{PWM} = \frac{f_{CLK}}{(PSC + 1)(ARR + 1)}$$

$$Duty = \frac{CCR}{ARR + 1} \times 100\%$$

- Example: $f_{CLK} = 72 \text{ MHz}$, $PSC = 71$, $ARR = 999 \rightarrow f_{PWM} = 1 \text{ kHz}$
- $CCR = 500 \rightarrow Duty = 50\%$

Example: LED Dimming

- Configure TIM2 Channel 1 as PWM output
- Connect LED to PA0 (TIM2_CH1)
- Vary CCR1 from 0 to ARR
- Observe brightness change

Example: Servo Motor

- Typical servo uses 50 Hz (20 ms period)
- Pulse width 1–2 ms controls angle (0° – 180°)
- Configure timer for 20 ms period
- Adjust CCR for position control

Summary

- PWM is widely used in embedded systems
- STM32 timers provide flexible PWM generation
- ARR defines period (frequency), CCR defines duty cycle
- Applications: LED, motor, audio, servo

Questions?